

Homework: Sparse Group Lasso

จะได้ **Sparse Group Lasso Penalty** จากทั้ง 12 กลุ่มดังนี้

$$\min \text{MSE} + \alpha \lambda \sum_{k=1}^{30} |\beta_k| + (1 - \alpha) \lambda$$

$$[\sqrt{3} \sqrt{\beta_1^2 + \beta_2^2 + \beta_3^2} +$$

$$\sqrt{3} \sqrt{\beta_4^2 + \beta_5^2 + \beta_6^2} +$$

$$\sqrt{3} \sqrt{\beta_7^2 + \beta_8^2 + \beta_9^2} +$$

$$\sqrt{4} \sqrt{\beta_{10}^2 + \beta_{11}^2 + \beta_{12}^2 + \beta_{13}^2} +$$

$$\sqrt{5} \sqrt{\beta_{14}^2 + \beta_{15}^2 + \beta_{16}^2 + \beta_{17}^2 + \beta_{18}^2} +$$

$$\sqrt{2} \sqrt{\beta_{19}^2 + \beta_{20}^2} +$$

$$\sqrt{2} \sqrt{\beta_{21}^2 + \beta_{22}^2} +$$

$$\sqrt{3} \sqrt{\beta_{23}^2 + \beta_{24}^2 + \beta_{25}^2} +$$

$$\sqrt{2} \sqrt{\beta_{26}^2 + \beta_{27}^2} +$$

$$\sqrt{1} \sqrt{\beta_{28}^2} +$$

$$\sqrt{1} \sqrt{\beta_{29}^2} +$$

$$\sqrt{1} \sqrt{\beta_{30}^2}]$$